DHARAWATCH

SIH Final Demo Video Script

A page-by-page voice-over script where the **watershed is the hero**, with Evidence Review as the second main act. Built by reading the actual code in this repository, so every line in the script describes something the app really does.

Item	Value
Total length	4:30 maximum (11 scenes)
Spoken words	589 words, about 4:00 at a calm 140 words/min, leaving about 30 seconds for clicks and pauses
Hero	Narmada Basin (HydroBASINS id hybas-4050031610, level 5, ~96,271 km², curated demo)
Main features	1. Watershed page (all six panels) · 2. Evidence Review (intervention evidence dossier)
Page order	Home → Explore → Watershed → Evidence Review → Field → Compare → Ask → Evidence → back to Watershed
Live site needs	Earth Engine credentials (for all maps and metrics) and AI_API_KEY (for the AI briefs)

How to use this document

- Section 1 gives the story in one sentence and in 30 seconds.
- Section 2 is the **pre-recording checklist**. Some panels take 30–120 seconds to load, so pre-load them.
- Section 3 is the one-page **run sheet** to keep next to you while recording.
- Section 4 is the **full script**, page by page: clicks, what appears, the voice-over, a technical line for Q&A, and the transition.
- Sections 5–7 explain each watershed panel and Evidence Review in simple words, list what **not** to say or click, and give judge Q&A answers.

1. The story

Project in one sentence

DharaWatch lets anyone open any watershed on Earth, see its vegetation, water and moisture condition from live satellite data, spot what changed since last year, and check whether the interventions inside it are backed by real evidence.

Project in 30 seconds (about 75 words)

“Water problems don’t follow district lines, they follow watersheds. DharaWatch starts with the watershed. Pick any basin, like the Narmada, and its real boundary loads from the global HydroSHEDS dataset. Sentinel-2 satellite data is analysed live across that basin to show vegetation, surface water and moisture, how they changed since last year, and what needs attention. Then, for each check dam or plantation, we combine satellite, terrain and field photos into a clear evidence status, so officers can decide where to act.”

The problem

- Watershed programmes build check dams, farm ponds, percolation tanks and plantations, but checking their effect needs field visits, and progress is hard to verify across a whole basin.
- Satellite data is free, but it is raw. Officers need answers, not spectral bands.
- AI summaries can be untrustworthy. Decision-makers need to know where every number came from.

Why the watershed is the right unit

Rain falling anywhere in a watershed drains to the same outlet, so upstream land use, vegetation and structures directly affect downstream water. In DharaWatch everything is attached to the active watershed: fingerprint, layers, timeline, attention flags, the intelligence profile, interventions, field evidence and AI brief. Evidence Review always starts by choosing a watershed and then an intervention inside it.

Story arc used in the video

Act	Question the viewer hears	Screen
1. Problem	Is the watershed getting better?	Home
2. Enter	Where do I work?	Explore + sidebar
3. The hero	Which watershed? Show me its real boundary.	Watershed: Narmada
4. Condition	How healthy is it now?	Fingerprint + Layers
5. Change	How has it changed, and what needs attention?	Temporal Ribbon + Attention
6. Understanding	Tell me everything, in plain words, and don’t make anything up.	Watershed Intelligence + AI Brief
7. Proof	Did the intervention actually work?	Evidence Review
8. Loop	How does ground truth get in, and where is it stored?	Field · Compare · Ask · Evidence
9. Impact	Why does this matter?	Back to the watershed

Who uses it and what they decide

- **District / watershed officers**: which sub-basin or structure to inspect first (Attention flags, evidence status NEEDS UPDATE or INCONSISTENT).

- **Programme reviewers / auditors**: whether an intervention's recorded status (e.g. OPERATIONAL) is supported by field and satellite evidence, then MARK REVIEWED.
- **Field teams**: upload geo-tagged photos that automatically link to the watershed and nearby interventions.

2. Before you record (very important)

Earth Engine computations run live and are cached on the server for about 15 minutes. Open everything once, **within 15 minutes before recording**, so that in the video every panel loads instantly.

#	Do this	Why
1	Open /watershed and select Narmada Basin. Wait until Fingerprint shows 3/3 LIVE METRICS, Timeline shows DATES, Monthly indices load, and Attention shows its table.	First computation over a 96,000 km² basin can take 30–90 s.
2	Switch on NDVI once and wait for ACTIVE, then RESET LAYERS.	Warms the tile cache.
3	Open Watershed Intelligence → AI BRIEF → GENERATE. Make sure it says AVAILABLE. If it says REJECTED or UNAVAILABLE, press REGENERATE / RETRY.	Gemini can take up to ~2 min or be overloaded (HTTP 503).
4	Open Evidence Review for “Sardar Sarovar Check Dam Network” and let SATELLITE CHANGE and TERRAIN finish. Note the status it gives.	So you know what status will appear on camera. Pick the intervention whose result tells the best story (see note below).
5	Open Compare from the Watershed action dock and let one comparison finish.	Compare runs two Earth Engine composites.
6	In Field, upload one real geo-tagged photo from the Narmada area beforehand and save it.	Then Field and Evidence pages already have content.
7	Browser: full screen, 100% zoom, 1920×1080, hide bookmarks, close other tabs.	Panels are positioned for desktop width.
8	Hard-refresh /watershed just before recording, so panel positions are default.	Clean layout.

Choosing the intervention for scene 9. The 5 seeded Narmada interventions are: Sardar Sarovar Check Dam Network (Check Dam, OPERATIONAL), Shoolpaneshwar Riparian Buffer (Plantation, OPERATIONAL), Navagam Percolation Tank (NEEDS_REPAIR), Narmada Canal Recharge Shafts (COMPLETED), Kevadia Farm Pond Cluster (Farm Pond, OPERATIONAL). The evidence status is computed live, so **pre-run them and use the one with the clearest result**. If you use a different one, change the intervention name in the voice-over.

Do not click on camera (these are not wired up yet):

- **Explore search box**: pressing Enter always flies to a fixed simulated location (San Francisco), not the place you typed.
- **Explore LAYERS radio buttons and DATE / DATASET chips**: they change the highlight only; the map imagery does not change.
- **Home page numbers** (12.8 km², +21.7%, 96.8% confidence) are illustrative marketing examples. Do not say they are results.
- **Explore copilot / Ask numbers**: scene dates and metrics in these answers come from a simulated metadata generator, not Earth Engine. Show the feature, but do not quote its numbers.
- **Field “OVERALL AGREEMENT %”**: fixed percentages based on two thresholds. Show the label, do not present it as a measured accuracy.
- Monitor and Reports are hidden in the sidebar. Do not navigate to /watch or /reports.

3. Run sheet (keep this next to you)

Time	Page	Click	Say (first words)
0:00 – 0:25	Home Page	Start recording on the Home page, scrolled to the top (hero: “UNDERSTAND THE WATERSHED.”).	“Across India, money goes into check dams, farm ponds…”
0:25 – 0:45	Explore Page + Sidebar	Explore opens on a satellite map. Click once anywhere on land: the SELECTED LOCATION card appea…	“This is our Explore page: a live satellite map…”
0:45 – 1:05	Watershed: Open Narmada	On the welcome card, click the chip Narmada Basin (or top bar DEMO, then “Sardar Sarovar / Narm…	“This is the Watershed page, the heart of DharaWatch.…”
1:05 – 1:30	Watershed Fingerprint	Point the cursor at the Fingerprint panel. Wait until the pill shows 3/3 LIVE METRICS.	“Top left is the Watershed Fingerprint: the basin’s condition…”
1:30 – 1:55	Geospatial Layers	Click the NDVI row in the Fingerprint panel. This switches on the NDVI layer and opens the Laye…	“One click paints vegetation onto the map. The Geospatial…”
1:55 – 2:15	Temporal Ribbon	In the Temporal Ribbon, press Play. Dots light up one by one; the selected date, scene count an…	“Below is the Temporal Ribbon. Every dot is a…”
2:15 – 2:35	Areas Needing Attention	Point at the Attention panel: the message, any cards (e.g. VEGETATION DECLINE · MEDIUM) and the…	“Areas Needing Attention compares today with the same season…”
2:35 – 3:05	Watershed Intelligence + Ai Brief	Click the expand icon on WATERSHED INTELLIGENCE (it widens).	“Watershed Intelligence brings it all together: geography, sub-basins, river…”
3:05 – 3:50	Evidence Review (The Evidence Dossier)	On the Watershed top bar click INT (Intervention Passport). Five Narmada interventions are list…	“From the Intervention Passport I’ll review the Sardar Sarovar…”
3:50 – 4:15	Quick Tour: Field · Compare · Ask · Evidence	Sidebar → Field: show the 6-step stepper (Field Photo → GPS → Map → Satellite Window → Watershe…	“The rest of the sidebar feeds the same loop.…”
4:15 – 4:30	Closing	Click Watershed in the sidebar (or browser Back) so the Narmada boundary fills the screen. Hold…	“DharaWatch puts the watershed at the centre, from satellite…”

Scene	Words	Seconds available	Speaking time at 140 wpm
1. Home Page	56	25	24 s
2. Explore Page + Sidebar	48	20	21 s
3. Watershed: Open Narmada	57	20	24 s
4. Watershed Fingerprint	60	25	26 s
5. Geospatial Layers	49	25	21 s
6. Temporal Ribbon	41	20	18 s
7. Areas Needing Attention	48	20	21 s
8. Watershed Intelligence + Ai Brief	58	30	25 s
9. Evidence Review (The Evidence Dossier)	91	45	39 s
10. Quick Tour: Field · Compare · Ask · Evidence	58	25	25 s
11. Closing	23	15	10 s
TOTAL	589	270	252 s

If you run long: cut the optional clicks first, then shorten scene 10 to Field + Evidence only. Never cut scenes 3, 4, 8 or 9.

4. Full script, page by page

Each scene: **clicks** (exact sequence), **on screen** (what the viewer sees), **voice-over** (read this aloud), **tech line** (for Q&A or a caption; not read aloud) and the **transition**.

SCENE 1 · HOME PAGE	0:00 – 0:25
Route	/
Goal	Name the problem and make the watershed the hero from the first second.
On screen	Hero headline, Sentinel-2 hero image, pipeline section (Ask, Analyse, Verify).

Clicks

1. Start recording on the Home page, scrolled to the top (hero: “UNDERSTAND THE WATERSHED.”).
2. Slowly scroll once past “FROM SATELLITE TO WATERSHED INSIGHT” and “BUILT FOR ACCURACY” (Ask · Analyse · Verify).
3. Scroll back up and click **Workspace** (top-right).

VOICE-OVER (56 words)

“Across India, money goes into check dams, farm ponds and plantations. But it is very hard to see whether a watershed is actually getting better. That is the problem DharaWatch solves. As our home page says: understand the watershed. We turn free satellite data into watershed intelligence, and we connect it to evidence from the ground.”

Transition: “Let’s step into the workspace.”

SCENE 2 · EXPLORE PAGE + SIDEBAR	0:25 – 0:45
Route	/explore
Goal	Show it is a real map workspace, and use the sidebar to introduce every tool in one breath.
On screen	Full-screen satellite map, the location card, the expanded sidebar with labels.

Clicks

1. Explore opens on a satellite map. Click once anywhere on land: the SELECTED LOCATION card appears.
2. Hover the left sidebar so it expands and the labels show: Explore, Watershed, Field, Evidence Review, Ask, Compare, Evidence, Settings.
3. Click **Watershed** in the sidebar.

VOICE-OVER (48 words)

“This is our Explore page: a live satellite map where you can click any point to inspect it. On the left is our sidebar: Watershed, Field, Evidence Review, Ask, Compare and Evidence. Every one of these tools is built around one unit: the watershed. So let’s open one.”

Transition: “So let’s open one.” (click Watershed)

SCENE 3 · WATERSHED: OPEN NARMADA	0:45 – 1:05
Route	/watershed
Goal	The hero moment: a real basin boundary appears from a global dataset.
On screen	Blue Narmada boundary on the satellite map. Panels open around it: Fingerprint, Layers, Intelligence, Attention, Temporal Ribbon.

Clicks

1. On the welcome card, click the chip **Narmada Basin** (or top bar **DEMO**, then "Sardar Sarovar / Narmada").
2. The map flies to the basin and the blue boundary is drawn. The top bar reads: *Narmada Basin · WATERSHED · L5 · ~96,271 km² · HYDROSHEDS · DEMO · LIVE*.
3. Optional (2 sec): hover the **NEW** menu to show "Draw boundary on map".

VOICE-OVER (57 words)

"This is the Watershed page, the heart of DharaWatch. I'll open the Narmada basin, which drains to the Sardar Sarovar dam. This boundary is not hand-drawn. It is loaded live from HydroSHEDS, the global watershed dataset, through Google Earth Engine. You can also search any river, double-click anywhere on Earth, or draw your own area."

Tech line (Q&A / optional caption): HydroBASINS levels 3–8 are queried in Earth Engine. A point or search result becomes the full basin hierarchy, and the basin is named from the HydroSHEDS river network. Code: **server/geospatial.js** (resolvePoint, ensureSeeds), **src/pages/Watershed/Watershed.jsx** (activateContext).

Transition: "First, how healthy is it?"

SCENE 4 · WATERSHED FINGERPRINT	1:05 – 1:30
Route	/watershed (FIN panel, top-left)
Goal	Three numbers that describe the basin's condition, each showing where it came from.
On screen	NDVI (vegetation), NDWI (surface water), NDMI (vegetation moisture) with bars, a land-cover bar and drainage reaches/km.

Clicks

1. Point the cursor at the Fingerprint panel. Wait until the pill shows **3/3 LIVE METRICS**.
2. Hover one metric so the provenance line is readable (dataset · dates · scenes · formula · scale).

VOICE-OVER (60 words)

"Top left is the Watershed Fingerprint: the basin's condition in three numbers. NDVI for vegetation, NDWI for surface water, NDMI for vegetation moisture. Each is computed now from cloud-masked Sentinel-2 images across the whole basin, and each shows its source: the dates, the number of scenes and the formula. Below that is land cover and the river network."

Tech line (Q&A / optional caption): Sentinel-2 SR Harmonized, SCL cloud mask, median composite over the last 30 days (90 or 180 if needed), mean over the basin polygon. Land cover = Dynamic World mode. Drainage = HydroSHEDS FreeFlowingRivers. Code: computeFingerprint() in server/geospatial.js.

Transition: "Numbers are good, but let's see them on the map."

SCENE 5 · GEOSPATIAL LAYERS	1:30 – 1:55
Route	/watershed (LAY panel, bottom-left)
Goal	The basin lights up with real Earth Engine imagery.
On screen	Red-to-green NDVI raster over the basin, with a legend and a provenance line (dataset · window · scenes · tiles rendered).

Clicks

1. Click the **NDVI** row in the Fingerprint panel. This switches on the NDVI layer and opens the Layers panel.
2. Wait for the pill to say ACTIVE (tiles stream in). Drag the OPACITY slider once.
3. Optional: switch on **Land Cover · Dynamic World**, then click RESET LAYERS.

VOICE-OVER (49 words)

"One click paints vegetation onto the map. The Geospatial Layers panel has nine layers: true colour, vegetation, water, moisture, Dynamic World land cover, SRTM terrain, NASA SMAP soil moisture and the drainage network. All of them are real Earth Engine tiles over this watershed, each with its own legend."

Tech line (Q&A / optional caption): One layer registry (src/shared/layerRegistry.js) defines both the Earth Engine palette and the legend, so a legend can never disagree with the pixels. Tiles come from /api/analysis/layers.

Transition: "That's today. What about over time?"

SCENE 6 · TEMPORAL RIBBON	1:55 – 2:15
Route	/watershed (TIM panel, bottom-centre)
Goal	Show the time dimension: real satellite passes and monthly trends.
On screen	A ribbon of acquisition dots (blue = under 30% cloud), the header "N dates · N scenes", a 12-month chart and a table.

Clicks

1. In the Temporal Ribbon, press **Play**. Dots light up one by one; the selected date, scene count and cloud % appear.
2. Point at the MONTHLY INDICES chart (green NDVI, blue NDWI, cyan NDMI).

VOICE-OVER (41 words)

"Below is the Temporal Ribbon. Every dot is a real Sentinel-2 pass over the basin; blue means less than thirty percent cloud. Press play to step through time. The chart shows how vegetation, water and moisture changed month by month."

Tech line (Q&A / optional caption): Dates = metadata of every Sentinel-2 scene intersecting the basin (24 months for small basins, fewer for large ones). Monthly indices = one cloud-masked median composite per month. Note: playing the ribbon selects dates; it does not change the map layer.

Transition: "But what should an officer look at first?"

SCENE 7 · AREAS NEEDING ATTENTION	2:15 – 2:35
Route	/watershed (ATT panel, bottom-right)
Goal	Automatic change flags: the system tells you what changed.
On screen	Either flagged cards with “now / ref / Δ”, or “No indicator deviates…” plus the comparison table including SMAP soil moisture.

Clicks

1. Point at the Attention panel: the message, any cards (e.g. VEGETATION DECLINE · MEDIUM) and the table Indicator / Now / Year ago / Δ.
2. Optional: click **COMPARE** on a card (it opens Compare; come back with the browser Back button).

VOICE-OVER (48 words)

“Areas Needing Attention compares today with the same season last year. If any index moves by more than 0.1, or soil moisture drops, it is flagged with the exact before and after values. Comparing the same season means a normal dry summer is not mistaken for damage.”

Tech line (Q&A / optional caption): Rule-based, not ML: $|\Delta| \geq 0.10$ gives MEDIUM, ≥ 0.20 gives HIGH. SMAP soil moisture $\Delta \leq -0.05$ m³/m³ is flagged. Code: computeAttention() in server/geospatial.js. These flags are basin-wide, not pixel hotspots.
Transition: “Now let’s bring everything about this watershed together.”

SCENE 8 · WATERSHED INTELLIGENCE + AI BRIEF	2:35 – 3:05
Route	/watershed (DET panel, top-right)
Goal	One profile for the basin, and an AI summary that cannot make up numbers.
On screen	Geography (area, countries, sub-basins), Hydrology (outlet discharge, degree of regulation), land-cover bars, Wikipedia history, Commons photos, sources, AI brief.

Clicks

1. Click the expand icon on WATERSHED INTELLIGENCE (it widens).
2. Click the section tabs quickly: GEOGRAPHY → HYDROLOGY → LAND COVER → MEDIA.
3. Click **AI BRIEF**. **Pre-generate it before recording** (it can take up to ~2 min). Show the sections and the line “All numbers in the brief were checked against the supplied facts.”

VOICE-OVER (58 words)

“Watershed Intelligence brings it all together: geography, sub-basins, river discharge, how regulated the river is by dams, land cover, history, photos and sources. At the bottom is the AI Brief. Gemini writes the summary, but every number it writes is checked against our verified data. If the AI makes up even one number, the brief is withheld.”

Tech line (Q&A / optional caption): getBrief() in server/watershedIntel.js sends only a JSON of verified facts, then extracts every number in the reply and rejects the brief if any number is not in those facts (status REJECTED).
Transition: “The satellite tells us what changed. But did the work on the ground actually happen?”

SCENE 9 · EVIDENCE REVIEW (THE EVIDENCE DOSSIER)	3:05 – 3:50
Route	/evidence-review?watershed=hybas-4050031610&intervention;=…
Goal	Main feature #2: check one intervention against three independent sources and reach a decision.
On screen	A map around the site, before/after true-colour images, index table with open-water hectares, land-cover transition, terrain cards, the categorical status with reasons.

Clicks

1. On the Watershed top bar click **INT** (Intervention Passport). Five Narmada interventions are listed.
2. On **Sardar Sarovar Check Dam Network** click **REVIEW EVIDENCE**. (Or open it from the sidebar: Evidence Review, then pick watershed and intervention.)
3. Point at the header (WATERSHED · INTERVENTION · STATUS · ● LIVE DATA) and the BASELINE / CURRENT / RADIUS bar.
4. Scroll through: SATELLITE CHANGE (before/after Sentinel-2 images and NDVI/NDWI/NDMI table) → TERRAIN CONTEXT → SYNTHESIZED EVIDENCE ASSESSMENT.
5. Click **MARK REVIEWED**.

VOICE-OVER (91 words)

"From the Intervention Passport I'll review the Sardar Sarovar check dam network. Evidence Review is our evidence dossier for one structure. It brings together three independent sources: field photos and inspections, a Sentinel-2 before-and-after around the site, and terrain from SRTM and MERIT Hydro, showing the slope and how much water drains to this point. The result is not a made-up score. It is a clear status, like strong, needs update, limited or inconsistent, with the reasons. One click on Mark Reviewed saves it as a record."

Tech line (Q&A / optional caption): The baseline date is season-aligned (same day of year, the year before construction) to avoid seasonal false alarms. Analysis radius depends on structure type (e.g. check dam 50 m). Rules flag conflicts, e.g. a plantation marked OPERATIONAL while NDVI fell by 0.1 or more. Code: src/shared/evidenceStatus.js, computeSiteChange / computeSiteTerrain in server/geospatial.js.

Transition: "The rest of the sidebar feeds this same loop."

SCENE 10 · QUICK TOUR: FIELD · COMPARE · ASK · EVIDENCE	3:50 – 4:15
Route	/field, /compare, /ask, /evidence
Goal	Show each remaining tool for about 5 seconds, without running long jobs live.
On screen	Field stepper, the Compare split view, the Ask chat, and the Evidence list with the review record.

Clicks

1. Sidebar → **Field**: show the 6-step stepper (Field Photo → GPS → Map → Satellite Window → Watershed Context → Evidence Passport). Use a pre-uploaded observation if possible.
2. Sidebar → **Compare**: drag the Split slider once (pre-load a result before recording).
3. Sidebar → **Ask**: show the page and the Explorer / Expert toggle. Do not dwell on its numbers.
4. Sidebar → **Evidence**: click the newest INTERVENTION REVIEW record you just saved.

VOICE-OVER (58 words)

"The rest of the sidebar feeds the same loop. Field turns a geo-tagged photo into evidence: it reads the GPS, fingerprints the file, and pulls satellite context for that spot. Compare shows two dates side by side. Ask lets anyone question the map in plain language. And Evidence keeps every record, including the review we just saved."

Tech line (Q&A / optional caption): Field: EXIF parsing, SHA-256 photo hash, Gemini vision (deterministic fallback if no key), Sentinel-2 indices at the point. Compare: Earth Engine composites for both dates, with Split, Difference and Flicker modes.

Transition: "So what does this give a decision-maker?"

SCENE 11 · CLOSING	4:15 – 4:30
Route	Back to /watershed (Narmada boundary on screen)
Goal	Finish on the hero image and one memorable line.
On screen	Narmada basin boundary with the NDVI layer on.

Clicks

1. Click **Watershed** in the sidebar (or browser Back) so the Narmada boundary fills the screen. Hold.

VOICE-OVER (23 words)

“DharaWatch puts the watershed at the centre, from satellite to field to decision, and every number shows where it came from. Thank you.”

Transition: End.

5. The watershed, panel by panel (simple explanations)

The top bar has six toggle buttons (FIN, LAY, DET, ATT, INT, TIM), one per panel. Panels can be dragged, collapsed and closed. Every panel shows a status pill (LIVE, LOADING, PARTIAL, NO DATA, ERROR) instead of fake numbers.

Panel	What it shows	Where the data comes from	Say it simply
Command bar	Search (OpenStreetMap + Wikidata), active watershed name, level, area, source, LIVE pill, NEW (draw / import GeoJSON), DEMO (8 basins), SAVED	HydroSHEDS HydroBASINS via Earth Engine; saved watersheds in SQLite	“Pick any watershed on Earth by search, click or drawing.”
FIN · Watershed Fingerprint	NDVI, NDWI, NDMI with value bar and provenance; land-cover bar; drainage reaches and km	Sentinel-2 SR (last 30/90/180 days, cloud-masked median, basin mean); Dynamic World; HydroSHEDS rivers	“The basin’s condition in three numbers: green, water, moisture.”
LAY · Geospatial Layers	9 layers in 4 groups (Watershed, Imagery, Indices, Land) with on/off, opacity, legend, status	Earth Engine tile URLs; SRTM; NASA SMAP L4 (~11 km); Dynamic World	“Put any layer onto the watershed with one click.”
TIM · Temporal Ribbon	Every Sentinel-2 acquisition date (cloud %), play/step, monthly NDVI/NDWI/NDMI chart and table	Sentinel-2 scene metadata; monthly composites (up to 12 months)	“Every dot is a real satellite pass. The chart is the basin’s pulse.”
ATT · Areas Needing Attention	Flag cards (vegetation decline, water recession, etc.) with severity; table now vs year ago vs Δ	Same window one year earlier; thresholds ±0.10 index, −0.05 m³/m³ SMAP	“It tells you what changed versus the same season last year.”
DET · Watershed Intelligence	12 sections: Overview, Geography, Hydrology, Earth Observation, Land Cover, Temporal Change, Interventions, Field Evidence, History, Media, Sources, AI Brief	HydroSHEDS (Pfafstetter, upstream area, discharge, regulation), LSIB countries, Nominatim, Wikipedia, Wikidata, Wikimedia Commons, Gemini	“Everything about this watershed in one profile, with sources.”
INT · Intervention Passport	Interventions in this watershed: type, status, coordinates, inspections; add / edit / inspect / delete; REVIEW EVIDENCE	SQLite interventions table (5 Narmada demo interventions)	“Every check dam, pond and plantation, with its own history.”
Action dock (bottom)	EXPLORE · COMPARE · FIELD · EVIDENCE REVIEW · DETAILS · ASK; the active watershed is passed along	Router state	“The watershed travels with you into every tool.”

The three indices, explained for judges

Index	Formula (Sentinel-2 bands)	Plain meaning	Why it matters for a watershed
NDVI	(B8 − B4) / (B8 + B4)	How green and dense vegetation is	Vegetation holds soil and slows runoff. A fall can mean degradation or drought.
NDWI	(B3 − B8) / (B3 + B8)	Open-water signal (above 0 is usually water)	Shows surface water in rivers, ponds and reservoirs. A basin-wide average is normally negative.
NDMI	(B8 − B11) / (B8 + B11)	Water content of vegetation (not soil)	Early sign of water stress before plants visibly brown.

6. Evidence Review: the evidence dossier

Route **/evidence-review**. Header badge: “INTERVENTION EVIDENCE REVIEW · DHARAWATCH · SIH26015”. Flow: **Watershed → Intervention → three evidence sources → categorical status with reasons → AI brief → MARK REVIEWED → Evidence page record.**

Part	What it does	Real data / logic
Context bar	Select WATERSHED, INTERVENTION, BASELINE date, CURRENT date, RADIUS; ANALYZE / RE-RUN	Default baseline = same day of year, the year before construction (season-aligned). Radius by type: check dam 50 m, percolation tank 100 m, recharge structure 25 m…
Field Evidence	Photos linked to the intervention or within 1 km, inspection records, latest date, photo viewer with EXIF and AI observation	field_observations + interventions tables. “Fresh” = under 365 days.
Satellite Change	Before/after Sentinel-2 true-colour images; NDVI/NDWI/NDMI before → after → Δ; open-water hectares; Dynamic World land-cover transition; Observation vs Interpretation text	Sentinel-2 SR, ±45 days around each date, requires 80% clear pixels in the footprint. Text says the change “does not by itself establish that the intervention caused the change”.
Terrain Context	Elevation, slope, flow accumulation level (LOW / MODERATE / HIGH / VERY HIGH), distance to the nearest mapped river	SRTM 30 m, MERIT Hydro upstream area, HydroSHEDS rivers
Timeline	Construction, inspections, photos, satellite scenes, previous reviews in date order	Built from the records above
Synthesized Evidence Assessment	STRONG EVIDENCE / NEEDS UPDATE / LIMITED EVIDENCE / INCONSISTENT / NO DATA, with reasons and limitations	Rule-based in src/shared/evidenceStatus.js. Deliberately no numeric score. Conflicts: e.g. field says “PARTIAL BREACH” while the record says OPERATIONAL, or plantation NDVI fell by 0.1 or more.
AI Evidence Brief	Plain-language summary of the evidence	Gemini on the server, numbers checked against the facts
Decision / Action	MARK REVIEWED	Saves the outcome on the intervention and creates an evidence record shown at /evidence

Other pages (for the quick tour and Q&A)

Page	What is real	Demo note
Home /	Navigation; the EXPLORE WATERSHED and ASK DHARAWATCH buttons	Section content and numbers are illustrative.
Explore /explore	Satellite basemap, click → location card, locate-me, AI copilot panel (calls /api/ai/ask), sidebar	Search, layer radios and date chip are not wired. Copilot metrics are simulated.
Field /field	Photo upload, SHA-256 hash, EXIF GPS/camera, Gemini vision (fallback without key), Google map location, Sentinel-2 indices at the point, watershed resolution, evidence passport saved to the DB	Needs the Google Maps key for the map step. Agreement % is heuristic.
Compare /compare	Earth Engine cloud-free composites for two dates; Split / Difference / Flicker; NDVI, NDWI and spectral-change metrics	Pre-load before recording.
Ask /ask	Chat UI, Explorer/Expert modes, GeoTIFF upload, intent parsing, prompt-injection guard, query log in DB	Satellite metadata is simulated. Ask does not read the selected watershed’s data. Use the AI Brief to show grounded AI.
Evidence /evidence	All evidence records (field observations, intervention reviews, detections) with source, hash, location, reasons, limitations; OPEN IN EVIDENCE REVIEW	On Vercel, records reset when the function restarts.

7. Priorities, wow moment and judge Q&A

Priority	Features
MUST SHOW	#1 Watershed: Narmada boundary from HydroSHEDS, Fingerprint, Layers (NDVI), Temporal Ribbon, Attention, Watershed Intelligence + AI Brief · #2 Evidence Review with its categorical status
SHOULD SHOW	Home hero, Explore + sidebar tour, Intervention Passport, MARK REVIEWED → Evidence record
ONLY IF TIME PERMITS	Draw-a-polygon region analysis (lists every intersecting watershed by level), sub-basin chips (click to drill down), Field stepper, Compare split
DO NOT SHOW	Explore search, Explore layer radios, quoting Ask/copilot numbers, Field agreement %, Monitor, Reports, Detect, Measure, Area, Timeline pages (not in the sidebar)

STRONGEST WOW MOMENT

Clicking a demo chip, the map flies across India and the **real Narmada basin boundary** appears. Seconds later its NDVI, NDWI and NDMI tick to LIVE with dates and scene counts, and one click paints the whole basin in NDVI. **Second wow:** the AI Brief, with the line “every number the AI writes is checked against verified data, and if it makes one up, the brief is withheld”.

Judge value proposition

- **Watershed-first:** any basin on Earth, resolved live from a global hydrological dataset, not a hard-coded map.
- **Traceable:** every metric shows its dataset, date window, scene count, formula and scale; every panel shows a status instead of fake values.
- **Honest AI:** AI only summarises verified facts, and an automatic number check blocks made-up figures.
- **From sky to ground to decision:** satellite change, terrain and field photos are combined per intervention into a reviewable, saved decision.
- **Free and open data:** Sentinel-2, Dynamic World, SRTM, SMAP, HydroSHEDS, MERIT Hydro, all through Google Earth Engine.

Likely judge questions

Question	Answer (accurate to the code)
Is this real data or a mock?	Watershed boundaries, fingerprint, layers, timeline, attention, compare and evidence-review satellite/terrain values are computed live in Google Earth Engine. Panels show NO DATA or ERROR instead of inventing values.
How do you detect change?	We compare the current composite with the same date window one year earlier (same season), flag a change of 0.10 or more in an index (0.20 or more = HIGH), and a soil-moisture drop of 0.05 m^3/m^3 or more. It is rule-based and transparent, not a black-box model.
How do you handle clouds?	Sentinel-2 Scene Classification (SCL) masks cloud, shadow, cirrus and saturated pixels, then a median composite is used. Evidence Review needs 80% clear pixels within ±45 days.
Can the AI hallucinate?	The brief gets only a JSON of verified facts. Every number in its output is matched against those facts; any unknown number causes REJECTED and the brief is not shown.
Why no evidence score?	There is no defensible basis for a numeric score, so we give a categorical status (STRONG, NEEDS UPDATE, LIMITED, INCONSISTENT, NO DATA) with the reasons that produced it.
Does it scale?	Yes, any HydroBASINS level 3–8 basin worldwide, from about 3,700 km^2 Bhadar to the Amazon. Large basins take 30–90 s the first time; results are cached for 15 minutes.
Resolution limits?	Indices are 10–20 m. Small seasonal ponds can be below 10 m; SMAP is about 11 km, so it is marked coarse. Both limits are stated in the UI.
Tech stack?	React + Vite + MapLibre frontend; Node/Express API; Google Earth Engine; SQLite; Gemini; OpenStreetMap Nominatim, Wikidata, Wikipedia, Wikimedia Commons; deployed on Vercel.

8. Architecture and what is confirmed

USER → React + Vite frontend (MapLibre map, floating panels) → /api on Node/Express (server.js + server/geoRoutes.js) → Google Earth Engine (HydroBASINS, HydroSHEDS rivers, Sentinel-2 SR, Dynamic World, SRTM, SMAP, MERIT Hydro, LSIB) → processing (cloud mask, median composite, index maths, zonal mean, same-season comparison, site analysis) → SQLite (watersheds, interventions, field_observations, evidence, ai_queries…) → Gemini (AI brief with number check, photo vision) → results as JSON and Earth Engine tile URLs → panels, layers, charts, evidence records.

External: OpenStreetMap Nominatim (search / nearest place), Wikidata + Wikipedia + Wikimedia Commons (river facts, history, photos), Google satellite basemap, Google Maps (Field). Hosting: Vercel (vercel.json, api/index.js) or Netlify. No user login.

Key files

File	Controls
src/pages/Watershed/Watershed.jsx	Watershed page: map, search, demo/saved, draw, panels, action dock, active-watershed hand-off
src/pages/Watershed/WatershedPanels.jsx · DetailPanel.jsx	Fingerprint, Layers, Temporal Ribbon, Attention, Interventions, Watershed Intelligence and AI Brief UI
src/shared/layerRegistry.js	The 9 layers: datasets, palettes, legends
server/geospatial.js	resolvePoint, demo seeds, computeFingerprint, computeTimeline(+Indices), computeAttention, computeLayerTile, computeSiteChange, computeSiteTerrain
server/watershedIntel.js	Geography, hydrology, history, media, and getBrief (Gemini + number check)
server/geoRoutes.js	/api/watersheds/*, /api/analysis/* routes
src/pages/EvidenceReview/* · src/shared/evidenceStatus.js	Evidence Review UI and categorical status rules
server.js	Field, interventions, evidence, evidence-review, compare, ask APIs; SQLite helpers
server-gee.js	Compare: Earth Engine composites and difference maps

Confirmed vs not confirmed

Area	Status
Watershed boundary, search, demo, saved, draw, import	IMPLEMENTED (Earth Engine + SQLite)
Fingerprint, Layers, Temporal Ribbon, Attention, Intelligence	IMPLEMENTED, live Earth Engine
AI Brief (watershed) and AI Evidence Brief	IMPLEMENTED; needs AI_API_KEY; can be REJECTED or UNAVAILABLE
Evidence Review + MARK REVIEWED → Evidence	IMPLEMENTED
Field pipeline	IMPLEMENTED (vision falls back to deterministic output without a key); agreement % is heuristic
Compare	IMPLEMENTED (Earth Engine)
Ask / Explore copilot	PARTIAL: UI, intent parsing and logging are real; satellite metadata and metrics are simulated
Explore search, layer switcher, date chip	MOCK / not wired
Home page metrics and preview	ILLUSTRATIVE (static)
Monitor, Reports	Hidden: not demo-ready
Live-site values (actual NDVI, flags, evidence status)	NOT CONFIRMED FROM CODE: they depend on the day's Earth Engine results. Check during pre-recording.

Area	Status
Screen recording cross-check	No recording or screenshots were provided in this session, so the visible-vs-code table could not be built. Share them and I can add it.